J S 3 SECOND TERM EXAMINATION

SUBJECT: MATHEMATICS

INSTRUCTIONS: Answer all questions in section A and any five (5) in section B.

SECTION A: OBJECTIVES

1. Which of the following numbers is not a perfect square?
   A. 64
   B. 81
   C. 100
   D. 120
   ANSWER= D
2. Find the LCM of 30, 45, 60 and 90.
   A. 5
   B. 60
   C. 180
   D. 360
   ANSWER=C
3. Simplify $\frac{3}{100} + \frac{30}{100}$, correct to 1 decimal place.
   A. 0.2
   B. 0.3
   C. 0.4
   D. 0.7
   ANSWER= B
4. Simplify $\frac{\sqrt{144}}{3}$
   A. 4
   B. 6
   C. 12
   D. 24
   ANSWER=A
5. Express 2211_{four} as denary number
   A. 48
   B. 160
   C. 165
   D. 168
   ANSWER=C
6. Find the square root of $\frac{750}{1080}$
   A. $\frac{2}{3}$
   B. $\frac{3}{4}$

C. $\frac{4}{5}$

D. $\frac{5}{6}$

ANSWER= D

7. Evaluate 5^3 - $\sqrt{25}$

A. 150

B. 125

C. 120

D. 0

ANSWER=C

8. How many vertices does a cuboids have?

A. 4

B. 6

C. 8

D. 12

ANSWER=C

9. Find the perimeter of a square whose side is 10cm

A. 10cm

B. 25cm

C. 35cm

D. 40cm

ANSWER= D

10. The Volume of a cylinder is given by the formula:

A. πr

B. πr^2

C. πr^2h

D. $2\pi r$

ANSWER= C

11. Find the radius of a circle, if its circumference is 132cm. (Take $\pi = \frac{22}{7}$)

A. 14cm

B. 16cm

C. 18cm

D. 21cm

ANSWER=D

12. Simplify 4(a+1) +5(a+2)

A. 5a + 10

B. 9 + 14a

C. 9a + 14

D. 10a + 14

ANSWER=C

13. Solve the equation $\frac{12}{3x+1} = \frac{3}{x}$.

A. 5
B. 4
C. 3
D. 1

ANSWER= D

14. Find the LCM of $9x\ and\ 4x^2y$

A. $9x^2y$
B. $18x^2y$
C. $36x^2y$
D. $45x^2y$

ANSWER=C

15. Solve the inequality $2x - 3 > 11$

A. x >3
B. $x > 7$
C. x >8
D. x >11

ANSWER= B

16. Solve the equation $\frac{x}{3} + x = 1$

A. 4
B. 3
C. 1
D. $\frac{3}{4}$

ANSWER=D

17. Expand and sim[plify (2x - 3)(x- 5)

A. $2x^2 - 13x - 15$
B. $2x^2 - 13x + 15$
C. $2x^2 + 13x - 15$
D. $2x^2 + 13x + 15$

ANSWER= B

18. Factorize $9a + 27$

A. 9(a+3)
B. 9(a - 27)
C. 9(a-3)
D. 9(3a- 7)

ANSWER=A

19. If the sum of the square of six and the square root of 81 is divided by 5, the result is

A. 11
B. 9
C. 5
D. 3

ANSWER= B

20. If -7 is added to a certain number, and the result is 45. What is the number?
    A. 42
    B. 35
    C. 52
    D. 12.

    ANSWER= C

21. Find the positive difference between $18\ and\ 43$
    A. -25
    B. -68
    C. 25
    D. -18

    ANSWER= C

22. The product of 8 and r is 480. Find r.
    A. 8
    B. 8r
    C. 480
    D. 60

    ANSWER=D

23. Write 0.00000082 in standard form
    A. 10×10^8
    B. 8.2×10^7
    C. 8.2×10^{-7}
    D. 8.2×10^{-8}

    ANSWER=C

24. Make t the subject from the formula $c = a + kt$
    A. $t = c + a - k$
    B. $t = \frac{c-a}{k}$
    C. $t = \frac{k-a}{c}$
    D. $t = \frac{a-k}{c}$

    ANSWER= B

25. If $y = \frac{f}{g} - h$. What is y when f =13, g = 26 and h=$\frac{3}{7}$?
    A. $\frac{1}{14}$
    B. $\frac{2}{14}$
    C. $\frac{3}{14}$
    D. $\frac{4}{14}$

    ANSWER= A

26. Evaluate $d^2 - 2u$ when d = 16 and u = 56.
    A. 144

B. 12
C. 256
D. 112
ANSWER= A

27. Calculate the simple interest on ₦4500 f0r 3 years at 6% per annum.
A. ₦810
B. ₦820
C. ₦830
D. ₦840
ANSWER=A

28. If $b = \frac{\frac{1}{2}c}{3d-4}$, find the value of b when c = 8, d =4.
A. 0.2
B. 0.3
C. 0.4
D. 0.5
ANSWER= D

29. Factorize $16xy + 14x^2$.
A. $2x(8y + 7x)$
B. $3y(8x + 7y)$
C. $4y(2x + 7y)$
D. $5y(3x + 3y)$

ANSWER= A

30. Expand and Simplify $2(a-2) + 3(a-3)$
A. $5a - 14$
B. $6a - 13$
C. $4a + 13$
D. $5a - 13$
ANSWER=D

31. What is the square of the square root of 81
A. 9
B. 900
C. 81
D. 91.
ANSWER=C

32. Find the square root of $\frac{75}{300}$
A. $\frac{1}{2}$
B. $\frac{1}{5}$
C. 2
D. 4

ANSWER=A

33. Simplify $\frac{9}{13} \div \frac{3}{4}$

A. $\frac{13}{12}$

B. $\frac{28}{39}$

C. $\frac{12}{13}$

D. $\frac{52}{28}$

ANSWER=C

34. Simplify $\frac{2}{3} + \frac{1}{5}$

A. $\frac{13}{15}$

B. $\frac{15}{13}$

C. $\frac{17}{15}$

D. $\frac{10}{15}$

ANSWER=A

35. Divide 100 by the positive difference between 65 and 75.

A. 5

B. 10

C. 15

D. 20

ANSWER= B

36. Convert $9\frac{1}{5}$ to improper fraction.

A. $\frac{9}{5}$

B. $\frac{10}{7}$

C. $\frac{46}{5}$

D. $\frac{63}{7}$

37. If 9 packets of cream cost ₦4500.00. Find the cost of 6 packets.

A. ₦2000.00

B. ₦2150.00

C. ₦2250.00

D. ₦3000.00

ANSWER=D

38. A die is rolled, what is the probability of getting a perfect square

A. $\frac{1}{3}$

B. $\frac{1}{2}$

C. $\frac{2}{3}$

D. $\frac{1}{6}$

ANSWER=A

39. How many edges does a cube have?
    A. 12
    B. 10
    C. 8
    D. 6

    ANSWER=A

40. How many vertices does a rectangular pyramid have?
    A. 4
    B. 5
    C. 6
    D. 7

    ANSWER= B

41. Find the reciprocal of $\frac{5}{6}$
    A. 1.5
    B. 1.2
    C. 3.5
    D. 4.5

42. Express 0.35 as a fraction in the lowest terms.
    A. $\frac{20}{7}$
    B. $\frac{7}{20}$
    C. $\frac{5}{7}$
    D. $\frac{7}{5}$

    ANSWER=B

43. Find the reciprocal of 8.
    A. 0.125
    B. 1.125
    C. 2.125
    D. 3.125

    ANSWER=A

44. Evaluate $1010_{two} - 101_{two}$
    A. 111_{two}
    B. 110_{two}
    C. 101_{two}
    D. 1011_{two}

    ANSWER=C

45. 5 pies cost ₦730. How much will 9 pies cost?
    A. ₦1,314
    B. ₦1,214

C. ₦1,414
D. ₦1,514
ANSWER=A

46. Find the value of $(1.1_{two})^3$ in base 10.
A. $\frac{27}{8}$
B. $\frac{9}{4}$
C. $\frac{1}{27}$
D. $\frac{1}{8}$
ANSWER=A

47. Convert 25_{ten} in base two.
A. 11001_2
B. 11010_2
C. 10011_2
D. 1001_2
ANSWER= A

48. Find the area of a square with length of sides 0.5m
A. $25m^2$
B. $0.2m^2$
C. $0.25m^2$
D. $0.52m^2$
ANSWER= C

49. 12 nights camp accommodation cost ₦6720. What will one night cost?
A. ₦560
B. ₦460
C. ₦360
D. ₦660
ANSWER=A

50. The base (b) of a triangular object is 9cm and the height (h) is 6cm. Find its area.
A. $27cm^2$
B. $54cm^2$
C. $28cm^2$
D. $29cm^2$
ANSWER=A

51. Find the area of a circle of radius 7cm (Take $\pi = 22/7$)
A. $154cm^2$
B. $308cm^2$
C. $408cm^2$
D. $254cm^2$
ANSWER=A

52. The ratio of the circumference of a circle to its diameter is known as ---

A. $pi(\pi)$
B. $diameter$
C. Radii
D. Non-rational
ANSWER=A

53. Evaluate $101_2 \times 11_2$
A. 1111_2
B. 1001_2
C. 1100_2
D. 10011_2

ANSWER=A

54. Solve the equation $\frac{2}{m-1}$ =1
A. 3
B. 4
C. 1
D. 2
ANSWER=A

55. Find the HCF of $15xy^2$ and $25x^2y$
A. 5xy
B. 10xy
C. 15xy
D. 25xy
ANSWER=A

56. Factorize $9x^2 - 1$
A. $9(x^2 - 1)$
B. $(3x^2 - 1)$
C. $(3x - 1)(3x + 1)$
D. $(3x + 1)(3x + 1)$

ANSWER=C

57. Make h the subject of formula $V = \pi h(R + r)$.
A. $h = \frac{r}{V(R\pi)}$
B. $h = \frac{V}{\pi(R+r)}$
C. $h = \frac{R}{V(r\pi)}$
D. $h = \frac{\pi}{V(R-r)}$

ANSWER= B

58. Multiply out the bracket -7(2a – 3b)

A. 7a + 2b
B. -7a – 21b
C. -14a + 21b
D. -7a + 21b
ANSWER=C

59. Find the square root of $\frac{100}{25}$
A. 2
B. 5
C. 10
D. 15
ANSWER=A

60. The diameter of a circle is $y + 2$. Calculate the radius of the circle?
A. $y + 2$
B. $\frac{y+2}{r}$
C. $\frac{y+2}{2}$
D. $\frac{y}{r}$
ANSWER= C

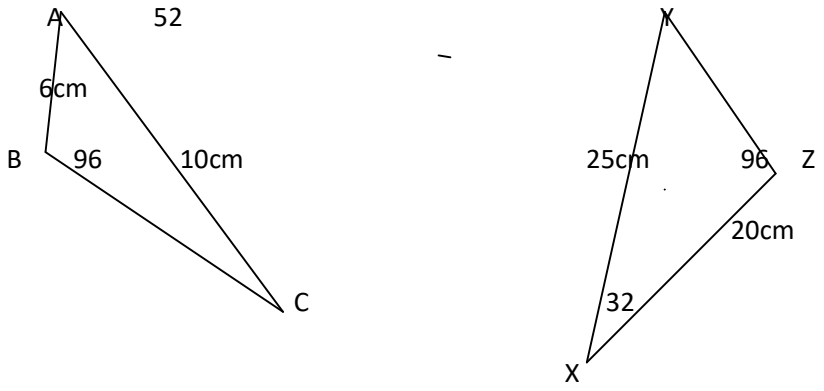
SECTION B: THEORY

1. Solve the equation $\frac{3x-2}{6} = \frac{4+5x}{5}$.
   a. Express34277_{eight} to a number in binary.
2. Calculate the 5th angle of the pentagon whose other four angles are $89°, 202°, 101°, and\ 132°$.
   b. The table below shows the scores of students in a mathematics an examination.

Scores	63	64	65	66	67
No. of students	6	4	3	5	2

   I. How many students took part in the test?
   II. What is the probability of scoring 64
   III. What is the probability of scoring 66
   IV. What is the probability of scoring 67.
   V. Calculate the percentage of scoring 63.
3. Make r the subject of formula $V = \pi h(R + r)$.
   I. If $= \frac{t+4u}{3t}$, find the value of f when $t = 10\ and\ u = 2$.

4. Name the similar triangles, giving the letters in corresponding order. Find the missing sides.



5. Find the compound interest and the final amount for ₦$40000\ for$ 2 years at 6% per annum.
6. The sketch the isometric view of the following three dimensional shapes with at least two properties (i) Cylinder (ii) rectangular pyramid (iii) Cuboid (iv)cone.